

Parag Jain

Email: jainparag.in@gmail.com

Web: parajain.github.io

Linkedin: www.linkedin.com/in/paragjain1

Google Scholar: scholar.google.com/citations/user=I6FyhVYAAAAAJ

Education

University of Edinburgh

Edinburgh, UK

CDT-NLP PhD, Natural Language Processing, Advisor: [Prof. Mirella Lapata](#)

Thesis: Explicit context representations for conversational language understanding

Fully funded by Huawei and UKRI.

Indian Institute of Technology

Hyderabad, India

M.Tech, Computer Science and Engineering, Advisor: [Prof. Vineeth N Balasubramanian](#)

Thesis: Metric Learning for Clustering in Streaming Large-Scale Data

Worked with [Dr. TVS Udaya Bhaskar](#) (INCOIS, India) on Ocean Biome Classification

Awarded A+ (highest possible) for research thesis

Jabalpur Engineering College

Jabalpur, India

B.E., Computer Science and Engineering

Experience

AI Research Scientist

Feb 2025 - Current

Meta Superintelligence Labs

Research Assistant

Feb 2024 - Dec 2024

University of Edinburgh

Google DeepMind

July 2023 - Nov 2023

Student Researcher

Tutor, University of Edinburgh

Jan 2021 - Apr 2021

Natural Language Understanding, Generation, and Machine Translation

IBM Research

Aug 2016 - Sept 2019

Research Engineer

Amazon

July 2015 - Aug 2016

Software Development Engineer

Teaching Assistant, IIT Hyderabad

July 2013 - June 2015

Numerical Linear Algebra, Database Management Systems, and Advanced Compiler Design

Conference Tutorial

Structured Data Summarization

Delivered a 3-hour tutorial at the Annual Meeting of the Association for Computational Linguistics (ACL), 2019, Florence (Italy).

Journal Publications

Memory-Based Semantic Parsing

Parag Jain, Mirella Lapata

Transactions of the Association for Computational Linguistics

2021

Scalable Micro-planned Generation of Discourse from Structured Data

Anirban Laha*, **Parag Jain***, Abhijit Mishra*, Karthik Sankaranarayanan

Computational Linguistics Journal

2019

Used for real-time verbal match insights during the spectator-free U.S. Open 2020, held without fans due to COVID-19.

In news: [\[US Open Blog\]](#) [\[Times of India\]](#) [\[Digital Creed\]](#)

Other Publications

Don't Waste Mistakes: Leveraging Negative RL-Groups via Confidence Reweighting

Yunzhen Feng, **Parag Jain**, Anthony Hartshorn, Julia Kempe, Yaqi Duan

Under Review at International Conference on Learning Representations (ICLR)

2025

What Characterizes Effective Reasoning? Revisiting Length, Review, and Structure of CoT

Yunzhen Feng, Julia Kempe, Cheng Zhang, **Parag Jain**, Anthony Hartshorn

Under Review at International Conference on Learning Representations (ICLR)

2025

Compute as Teacher: Turning Inference Compute Into Reference-Free Supervision Dulhan Jayalath, Shashwat Goel, Thomas Foster, Parag Jain , Suchin Gururangan, Cheng Zhang, Anirudh Goyal, Alan Schelten Under Review at International Conference on Learning Representations (ICLR)	2025
The llama 4 herd: The beginning of a new era of natively multimodal AI innovation Meta AI	2025
Integrating Large Language Models with Graph-based Reasoning for Conversational Question Answering Parag Jain , Mirella Lapata arXiv:2407.09506	2024
Structsum Generation for Faster Text Comprehension Parag Jain , Andreea Marzoca, Francesco Piccinno Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)	2024
Semantic Parsing for Conversational Question Answering over Knowledge Graphs Laura Perez-Beltrachini, Parag Jain , Emilio Monti, Mirella Lapata Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)	2023
Conversational Semantic Parsing using Dynamic Context Graphs Parag Jain , Mirella Lapata Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)	2023
Multi-Document Summarization with Centroid-Based Pretraining Ratish Puduppully, Parag Jain , Nancy F. Chen, Mark Steedman Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)	2022
Bootstrapping chatbot interfaces to databases Ashish Mittal, Diptikalyan Saha, Parag Jain , Jaydeep Sen, Manasa Jammi, Karthik Sankaranarayanan Proceedings of the 3 rd ACM India Joint International Conference on Data Science and Management of Data	2021
Unsupervised Controllable Text Formalization Parag Jain , Abhijit Mishra, Amar Prakash Azad, Karthik Sankaranarayanan Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)	2019
Storytelling from structured data and knowledge graphs: An NLG perspective Abhijit Mishra, Anirban Laha, Karthik Sankaranarayanan, Parag Jain , Saravanan Krishnan Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL, Tutorial Abstracts)	2019
Unified Semantic Parsing with Weak Supervision Priyanka Agrawal, Parag Jain , Ayushi Dalmia, Abhishek Bansal, Ashish Mittal, Karthik Sankaranarayanan Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)	2019
Unsupervised Neural Text Simplification Sai Surya, Abhijit Mishra, Anirban Laha, Parag Jain , Karthik Sankaranarayanan Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)	2018
A Mixed Hierarchical Attention Based Encoder-Decoder Approach for Standard Table Summarization Parag Jain , Anirban Laha, Karthik Sankaranarayanan, Preksha Nema, Mitesh M. Khapra, Shreyas Shetty Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)2018	
Generating Descriptions from Structured Data Using a Bifocal Attention Mechanism and Gated Orthogonalization Preksha Nema, Shreyas Shetty, Parag Jain , Anirban Laha, Karthik Sankaranarayanan, Mitesh M. Khapra Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL) 2018	
Story Generation from Sequence of Independent Short Descriptions Parag Jain , Priyanka Agrawal, Abhijit Mishra, Mohak Sukhwani, Anirban Laha, Karthik Sankaranarayanan Special Interest Group on Knowledge Discovery in Data (KDD)	2017
Creation and Interaction with Large-Scale Domain-Specific Knowledge Bases Shreyas Bharadwaj et al. [and others, including Parag Jain] Proceedings of the Very Large Data Bases Endowment (VLDB)	2017
Online Active Metric Learning for Clustering Parag Jain , Vineeth N. Balasubramanian Research Colloquium in the 3rd Xerox Research Centre India Open (XRCI Open)	2016
Patents	
Cognitive assistant for co-generating creative content – US Patent US20200134089A1 Anush Sankaran, Pranay Lohia, Priyanka Agrawal, Disha Shrivastava, Anirban Laha, Parag Jain	2022

Real-time assessment of text consistency – US Patent US11194964B2 Abhijit Mishra, Anirban Laha, Parag Jain, Karthik Sankaranarayanan	2021
Controllable Style-Based Text Transformation – US Patent US11734509B2 Abhijit Mishra, Parag Jain, Amar P. Azad, Karthik Sankaranarayanan	2021
Unsupervised Tunable Stylized Text Transformations – US Patent US11574132B2 Parag Jain, Amar P. Azad, Abhijit Mishra, Karthik Sankaranarayanan	2021
Structured article generation – US Patent US20210117431A1 Parag Jain, Abhijit Mishra, Anirban Laha, Saravanan Krishnan, Karthik Sankaranarayanan	2021
Script modification – US Patent US20200125638A1 Saneem Chemmengath, Parag Jain, Anirban Laha, Saravanan Krishnan	2021
Ontology-based automatic bootstrapping of state-based dialog systems – US Patent US20190228068A1 Jaydeep Sen, Parag Jain, Diptikalyan Saha, Ashish Mittal	2020
Generation of variable natural language descriptions from structured data – US Patent US20200073944A1 Abhijit Mishra, Parag Jain, Anirban Laha, Karthik Sankaranarayanan	2020
Displaying dynamic content on multiple devices – US Patent US10664217B1 Anirban Laha, Vijay Ekambaram, Parag Jain, Disha Shrivastava	2020
Simulating tactile information for haptic technology – US Patent US20190163272A1 Shreya Khare, Parag Jain, Srikanth G. Tamilselvam, Senthil Kumar Kumarasamy Mani, Sampath Dechu	2019

Invited Talks

- Generating and Reasoning over Structured Data Meta, Amazon, 2024
- Structsum Generation using Large Language Models IIT Hyderabad, 2024
- Foster Talk at Jabalpur Engineering College 2023
- Conversational Semantic Parsing - Online International Conference On Advances in Physical, Mathematical and Computational Sciences, India 2022
- Introduction to Machine Learning - Christ University Bangalore 2018
- Metric Learning for Clustering in Streaming Large-Scale Data - IBM Bangalore 2016
- Introduction to PageRank, IIT Hyderabad 2015

Awards

- Travel Award by Google DeepMind to present our work at ACL 2024
- Travel Award by Facebook AI Research to present our work at EurNLP 2019
- PhD Scholarship Award by Huawei 2019
- IBM Second Invention Plateau Award - 2019
- IBM First Invention Plateau Award - 2019
- Equity Award for outstanding achievement in research and development. - 2018
(**An individual award selected globally every year by the office of CEO of IBM.**)
- IBM Research First Patent Award - 2018
- Certificate of Appreciation from the Director of IBM Research India for my Contribution to Natural Language Generation Research in 2018
- Travel Grant by Xerox Research for attending XRCI Open, Bangalore - 2016
- Finalist Amazon Idea hackathon, Hyderabad 2016
- Travel Grant for attending the Fifth User Interaction Workshop at INCOIS, Hyderabad - 2015
- Ministry of Human Resource Development Scholarship by Government of India, 2013 - 2015
- Ministry of Human Resource Development Scholarship by Government of India, 2012

Mentorship

- Dulhan Jayalath (University of Oxford) – Research Scientist Intern, Meta 2025
- Yunzhen Feng (New York University) – Research Scientist Intern, Meta 2025
- Akash Kumar (IIT Madras) – Software Engineering intern, IBM 2018 (next PhD student at UCSD)
- Aman Sinha (IIT ISM), Software Engineering intern at IBM 2019 (next Research Assistant at IECL Laboratory, University of Lorraine)

Service

- Actively contribute as a reviewer for leading machine learning and NLP conferences such as COLM, ACL, EMNLP, and NAACL.
- Founding team member of TEDxIITHHyderabad, 2016
- Local coordinator for i.school workshop University of Tokyo, at IIT Hyderabad, 2015
- Student placement representative CSE Department IIT Hyderabad, 2015
- Core member of organizing team Aureoleâ€™10 Engineering Techfest, 2010